- Only one factor is variable while others are held constant.
- All units of the variable factor are homogeneous.
- The product is measured in physical units.
- There is no change in the state of technology.
- There is no change in the price of the product.

## Total Product (TP)

It refers to the total amount of commodity produced by the combination of all inputs in a given period of time.

- Summation of marginal products, i.e. **TP = ΣMP**

where, TP= Total Product, MP= Marginal Product

## Average Product (AP)

It is the result of the total product divided by the total units of the input employed. In other words, it refers to the output per unit of the input.

Mathematically, $AP = TP/N$

Where,

AP= Average Product

TP= Total Product

N= Total units of inputs employed

## Marginal Product (MP)

It is the addition or the increment made to the total product when one more unit of the variable input is employed. In other words, it is the ratio of the change in the total product to the change in the units of the input. It is expressed as

$$MP=\Delta TP/\Delta N$$

where,

MP = Marginal Product

ΔTP = Change in total product

ΔN = Change in units of input

It is also expressed as

$$MP = TP\ (n) - TP\ (n-1)$$

Where,

MP = Marginal Product

TP(n) = Total product of employing $n^{th}$ unit of a factor

TP(n-1) = Total product of employing the previous unit of a factor, that is, $(n-1)^{th}$ unit of a factor.

The Law of Variable Proportions is explained with the help of the following schedule and diagram:

In table 3.1, units of variable factor (labour) are employed along with other fixed factors of production. The table illustrates that there

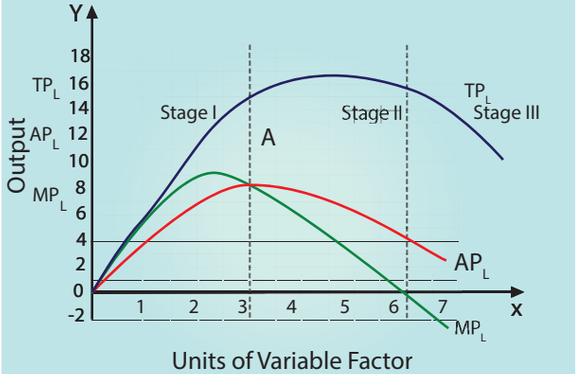



**Diagram 3.1**



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